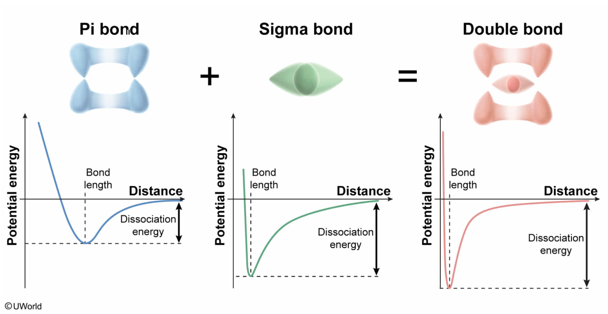
The C=S double bond in carbon disulfide (CS_2) consists of a combination of one σ bond and one π bond. Compared to the bond dissociation energy (bond strength) of the π bond portion of the C=S bond, the bond dissociation energy of the σ bond portion of the C=S bond is:

- A. greater. [63%]
- B. less. [34%]
- C. the same. [1%]
- D. proportional.

Correct
63% answered correctly

Explanation:



Covalent bonds between atoms are formed by sharing electrons, which is made possible through the overlap of the atomic orbitals from each atom. Covalent bonds made by the **end-to-end overlap** of atomic orbitals are called **sigma (σ) bonds** whereas covalent bonds made by the **side-to-side overlap** of *p* orbitals are called **pi (π) bonds**.

Although some atoms participate only in single covalent bonding (by σ bond), other atoms are capable of participating in **multiple bonds** (ie, a double or triple bond) by the addition of one or more π bonds. In this way, a double bond consists of one σ bond and one π bond, and a triple bond consists of one σ bond and *two* π bonds. Comparisons can be made between single, double, and triple covalent bonds provided that the bonds being compared involve the same two types of atoms.

The overall energy of a bond is the sum of the energies of all σ and π bonding contributors. The overall **bond dissociation energy** (ie, the *total energy* required to break both the σ bond and any π bonds) **increases** (relative to a single bond) with **each additional π bond**. Consequently, when comparing single, double, and triple covalent bonds involving the same two types of atoms, triple bonds are stronger and take more energy to break than double bonds, and double bonds are stronger and take more energy to break than single bonds.

However, if the strengths of the σ bond and π bond portions of a double or triple bond are considered *separately* and compared, the π bond portion is weaker than σ bond portion. Because the end-to-end orbital overlap in σ bonds is more efficient than the side-to-side orbital overlap in π bonds, σ bonds tend to exist in a **more stable, lower energy** state. As a result, breaking the σ bond portion requires more energy than breaking a π bond portion (ie, the σ bond portion has a **greater dissociation energy**).

(Choices B, C, and D) Due to differences in the efficiency and mode of the orbital overlap, the dissociation energies of the σ and π bond portions of a double bond are not equivalent or proportional, and breaking σ bonds requires *more* energy than breaking π bonds.

Educational objective:
Sigma bonds are lower in energy, more stable, and have a greater dissociation energy than π bonds. Although individual π bonds are weaker than σ bonds, double and triple bonds are composed of both σ and π bonds and are therefore stronger overall than a single bond.

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